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2. Geology and Crustal Structure of Svalbard and the Barents Sea

PALEOMAGNETIC INVESTIGATIONS OF MESOZOIC AND PALEOZOIC ROCKS FROM SVALBARD

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ABSTRACT

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A paleomagnetic study of Late Mesozoic dolerite dykes and sills and Paleozoic sediments from Spitsbergen, the main island of the Svalbard Archipelago, gives the position of the pole in the Late Mesozoic and Paleozoic as distinct from the corresponding poles of Europe and North America. The Paleozoic pole is to the south of corresponding poles for Europe and North America. The data suggest that Svalbard has moved independently of Europe and North America at least in the Late Mesozoic, and thus might behave as a microplate or block.

INTRODUCTION

The Svalbard Archipelago is now part of the Eurasia plate and became detached, together with Europe, from North America and Greenland during the Tertiary (Harland, 1969; Birkenmajer, 1972; Birkenmajer, 1981; Fig. 1a). Svalbard, lying at the northern end of the Mid-Atlantic Ridge, must have played an important part in the geotectonic history of the Arctic and North Atlantic Region. Paleomagnetic investigations in Svalbard may efficiently contribute to the reconstruction of the pre-Tertiary relative positions of North America and Eurasia.

Thus far, there are several paleomagnetic data from Svalbard (Krumsiek et al., 1966; Spall, 1968; Storetvedt, 1972; Halvorsen, 1972, 1973; Sandal and Halvorsen, 1973; Briseid and Halvorsen, 1974). Recent studies (Jeleńska et al., 1978; Vincenz et al., 1981; Vincenz and Jeleńska, 1984; Vincenz et al., 1984) have been undertaken in order to investigate the mode of opening of North Atlantic and breakup of Pangea.

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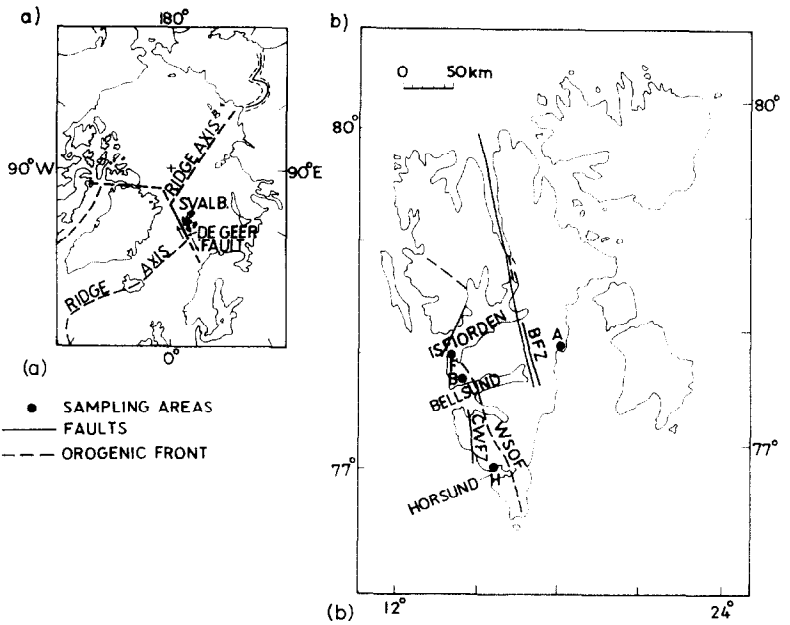


Fig. 1. Sketch map of Svalbard. a. Location of Svalbard in the Arctic region. b. Sampling areas: *A* — Agardhbukta sills; *H* — Hornsund dykes; *B* — Bellsund Paleozoic sediments; *F* — Festningen Paleozoic sediments; *WSOF* — West Spitsbergen orogenic front; *BFZ* — Billefjorden Fault Zone; *CWFZ* — Central West Fault Zone.

AREA OF STUDY AND SAMPLING MATERIAL

The initial study target was the Late Mesozoic dolerites in the form of dykes and sills but later sampling was extended to include Permian and Uppermost Carboniferous sediments.

Figure 1b shows a sketch map of Spitsbergen, the main island of Svalbard Archipelago and the positions of sampling localities. The Mesozoic dolerite dykes were sampled in the Hornsund region in the southwest (at *H*) and the corresponding sills at Agardhbukta on the east coast (*A*). The Permo-Carboniferous sediments were sampled at Festningen at the mouth of Isfjorden (*F*) and at Bellsund (*B*). Three of the areas investigated (*H*, *F* and *B*) lie within the Tertiary orogenic belt. The west Spitsbergen orogeny affected the western edge of Spitsbergen during Early to Mid-Tertiary times and was associated with the dextral displacement of Greenland (Harland, 1969; Birkenmajer, 1972; Birkenmajer, 1981). As a result of the orogeny all formations within the orogenic belt have been folded and subjected to regional hydrothermal alteration (Birkenmajer and Morawski, 1960).

THE MESOZOIC DOLERITE DYKES AND SILLS

The dolerite dykes sampled at Hornsund have been described by Birkenmajer and Morawski (1960) as being of Late Mesozoic age. The dykes are independent of the

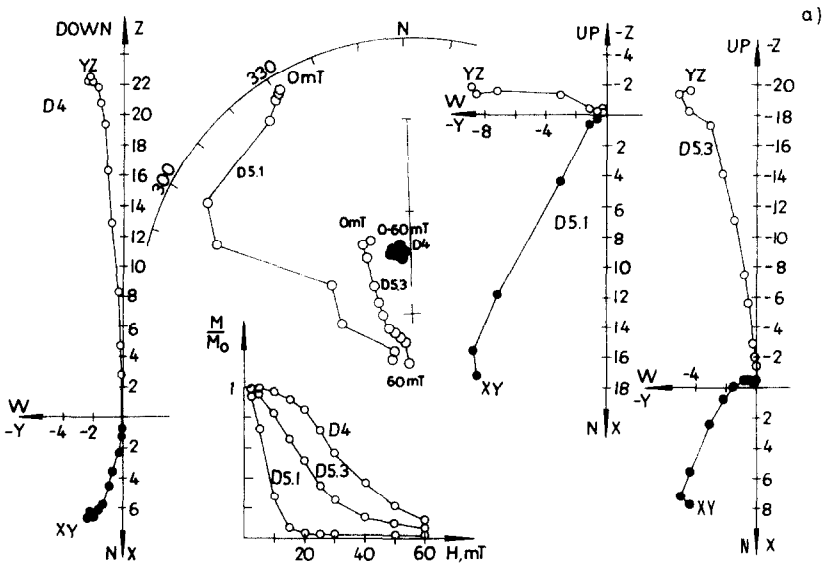
TABLE 1

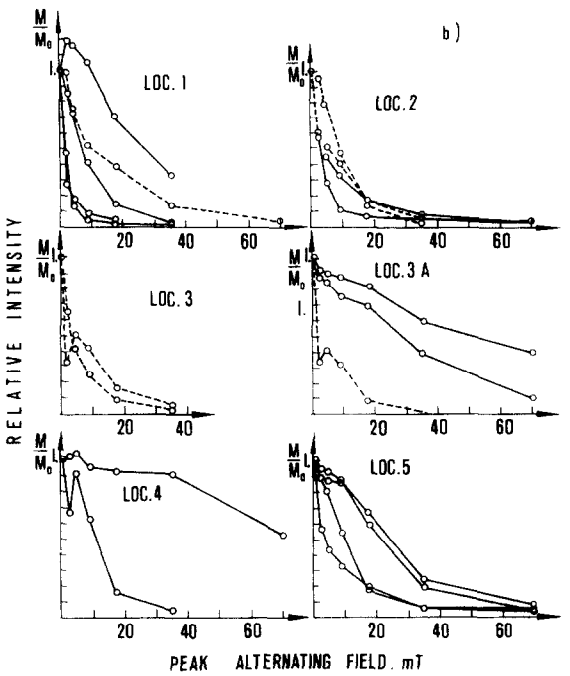
Comparison of conventional with the least squares analysis *

Dyke or sill	Conventional					Least squares analysis				
	n	dec. (°)	inc. (°)	k	α_{95} (°)	n	dec. (°)	inc. (°)	k	α_{95} (°)
D. 1	32	38.3	77.6	96.5	2.6	18	358.0	69.8	8.5	12.6
D. 2a	9	74.7	76.7	135.4	4.4	4	344.7	65.9	138.1	7.9
D. 2b	21	354.7	79.0	102.8	3.1	7	337.1	78.7	46.5	8.9
D. 3	36	0.1	77.7	206.9	1.7	13	8.3	73.4	56.5	5.6
D. 4	35	348.1	76.2	238.1	1.6	14	345.1	75.8	104.9	3.9
D. 5a	14	29.6	−68.6	146.0	3.3	8	7.6	−71.0	52.9	7.7
D. 5b	11	173.0	−79.9	98.3	4.6	5	143.6	−66.3	15.7	19.9
D. 6	24	170.4	−77.2	127.0	2.6	15	154.1	−76.1	45.0	5.8
S. 1	36	163.5	−47.8	14.7	16.3	37	169.4	−49.8	24.8	11.3
S. 2	38	177.5	−60.7	21.2	14.9	23	171.8	−63.7	20.9	12.4
S. 3	40	122.5	−59.2	29.8	12.5	20	134.3	−59.5	46.9	9.9
S. 3A	25	333.7	−72.5	218.1	8.4	8	325.0	+61.1	193.2	18.1
S. 4	57	170.6	−59.9	155.1	3.9	9	149.3	−67.4	105.0	6.6
S. 5	69	178.4	−61.9	44.3	7.3	7	184.0	−64.9	127.9	6.8
Mean for dykes	6	0.3	78.4	382.6	3.4	6	342.7	72.3	193.5	4.8
Mean for sills	5	162.6	−59.5	38.9	12.4	6	159.0	−62.2	55.7	9.0

* n is the number of specimens, dykes marked by a capital D, sills marked by a capital S; α_{95} confidence angle at the 95% probability level; k precision parameter. Dyke 7 which gives significantly different directions from the others, is omitted.

Caledonian structure of the region and have been intruded into the Precambrian sediments, metamorphics and granites, vertically or approximately vertically. The dykes comprise medium-to-fine-grained rocks, which contain titanomagnetite and





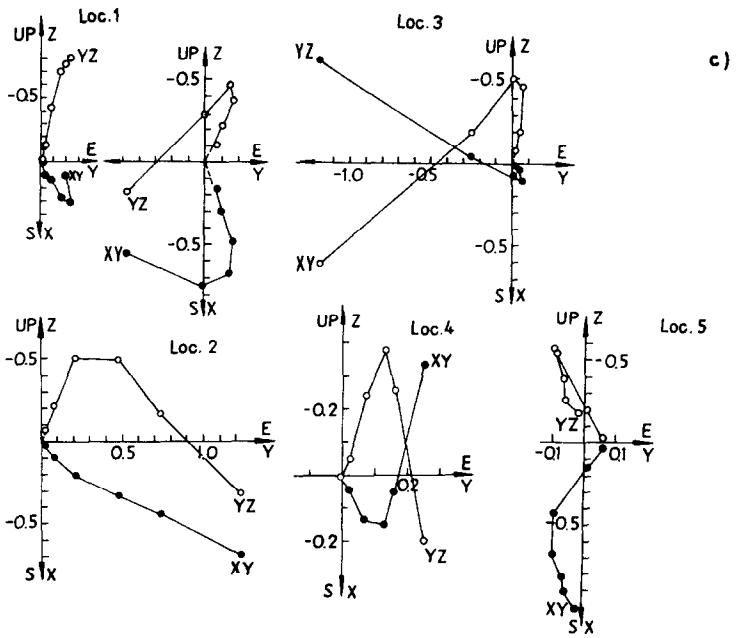


Fig. 2. Examples of AF-demagnetization curves, Zijderveld diagrams and directional changes for: (a) dolerite dykes; (b) and (c) dolerite sills. Units: 10^{-1} Am^{-1}; peak alternating fields in mT. Directional changes shown on stereonet: full circles lower hemisphere; open circles upper hemisphere.

ilmenite usually in the form of skeletal aggregates, and here and there accessory pyrite (Vincenz et al., 1981).

Microscopic studies of polished sections supported by transmitted light investigations showed that magnetic material in the dolerite sills consisted of medium-to-fine-grained titanomagnetite, relatively little altered by deuteric oxidation, with an overprint of hydrothermal changes, which are partly strong and partly moderate.

Detailed rock magnetic studies comprising thermomagnetic analysis, acquisition of TRM and IRM, the Lowrie-Fuller test (Lowrie and Fuller, 1971), the Hopkinson effect and microscopic analysis were performed (Jeleńska et al., 1978/79; Vincenz et al., 1984).

Paleomagnetic investigations of Late Mesozoic dolerites were described in detail by Vincenz et al. (1981) and Vincenz et al. (1984). The examples of demagnetization curves, Zijderveld diagrams and directional changes plotted on stereonet are shown in Fig. 2.

The directional changes such as illustrated in Fig. 2a show either a tightly grouped set of points (sample D.4) or a directional trend without a distinct end point. In the latter case, it was difficult to determine the final direction of remanence, which has been often achieved by using the linear decrease towards the origin on the Zijderveld plot.

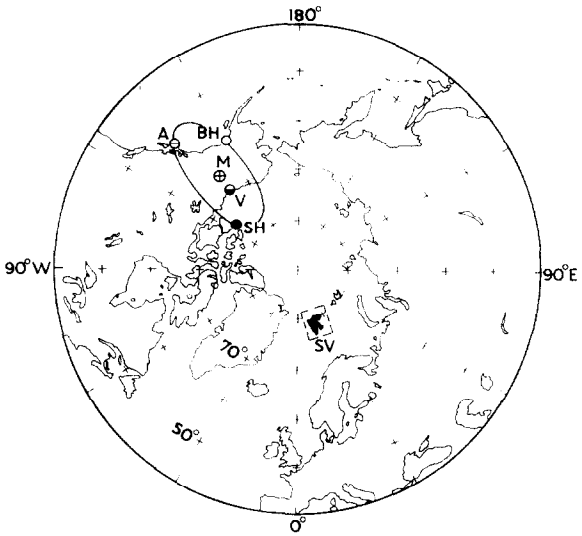


Fig. 3. Pole positions for Late Mesozoic dolerite of Spitsbergen: normal polarity: full circles; reversed polarity: open circles; mixed polarity: half-full circles. *SV* —Svalbard Archipelago; *A* —Agardhalen sills; *V* —Hornsund dykes; *SM* —Sandal and Halvorsen (1973); *BH* —Briseid and Halvorsen (1974); *M* —mean Late Mesozoic pole for Spitsbergen shown with circle of 95% confidence.

To check the validity of such procedure a least squares analysis of step demagnetization has been performed. The method developed by Stupavsky and Symons (1978) is based upon the assumption that the intensity of each component decays exponentially during the demagnetization process. The observed remanence vectors are tested for a best fit to the mathematical models, comprising one, two or three components of different direction, magnitude and stability. The results of the least squares analysis are summarized in Table 1.

Figure 3 shows the position of paleomagnetic poles based upon the data for the Late Mesozoic dolerites under study. Other available paleomagnetic poles for Spitsbergen were also included together with the position of the mean Mesozoic pole ($\lambda' = 65.2°N$; $\phi' = 221.7°E$; $\alpha_{95} = 12.3°$) which was calculated from our results and from the pole positions obtained by Sandal and Halvorsen (1973) and by Briseid and Halvorsen (1974).

PALEOZOIC SEDIMENTS

The Permian and Uppermost Carboniferous sediments were sampled at Festningen at the mouth of Isfjorden and at the mouth of Bellsund about 50 km to the south. Both areas lie within the Tertiary orogenic belt and have been folded (Fig. 1b).

The sampling horizons are indicated in Table 2 along the stratigraphic column. The dotted horizontal line within the Treskelodden formation shows that the

TABLE 2

An outline of the geologic section

System	Stage	Group	Formation	Other names	Paleomagnetic horizons	
					Festningen	Bellsund
Permian	Late Permian	Temple-fjorden	Kapp Starostin Fm	Brachiopod Cherty Lst. Fm.	Pd-8	P10-1 to 10
						P10-11 to 16
					P5-12 to 11	P10-17 to 19
				Spirifer Lst. Fm.	P5-10 to 7	P10-20
					Pd-7 to 6	P10-21
	Early Permian	Gipsdalen	Gipshuken Fm.	Treskelodden Fm.	P5-6 to P5-1	
			Norden-skiold-breen Fm.			
	Late				P4-5 to 8	
					P4-3 to 4	
Carbonif.	Middle	Billefjorden	Ebdalen Fm.	Hyrnefjellet Fm.	P4-2	Cd-7 to Cd-5
					P4-1	Cd-4 to Cd-3
					P4-9	
	Early		Svenbreen Fm.		P4-10	

boundary between the Late Carboniferous and Permian rocks is indistinct and not easy to define. There has been some argument among geologists where this boundary should be placed (Orvin, 1940; Forbes, 1960; Cutbill and Challinor, 1965; Harland, 1969; Birkenmajer, 1981).

Sampled rocks consist mainly of marly limestones, arenaceous limestones and cherty limestones (Siedlecka, 1970). All rocks are highly carbonaceous. Deposition of limestones under study occurred in uniform conditions, in a warm, arid climate, in a marine environment with eustatic changes of sea level (Birkenmajer, 1964). The magnetic mineralogy of limestones is difficult to describe optically because of the small grain size and the low concentration. According to Lowrie and Heller (1982) magnetic techniques are more promising. In the case of Spitsbergen limestones, extraction and concentration of the magnetic minerals failed. The only method is to observe the decay of saturation remanence I_{rs} during continuous heating in a field-free space. Figure 4 shows examples of typical decay curves for limestones from Festningen and Bellsund. In a fresh sample (Fig. 4a, curve *1*) the thermal behaviour of I_{rs}, revealed the presence of magnetite and or maghemite (sample P.10). In some cases, besides magnetite, the second phase of blocking temperature about 315°C is to be seen. After heating I_{rs} increased from ten up to a hundred fold and this increase is ascribed to the growth of new magnetic mineral. This mineral is seen on the decay curve of I_{rs} obtained during the second heating (Fig. 4, curves *2*). The blocking temperature of the new mineral is about 315°C and is typical of the

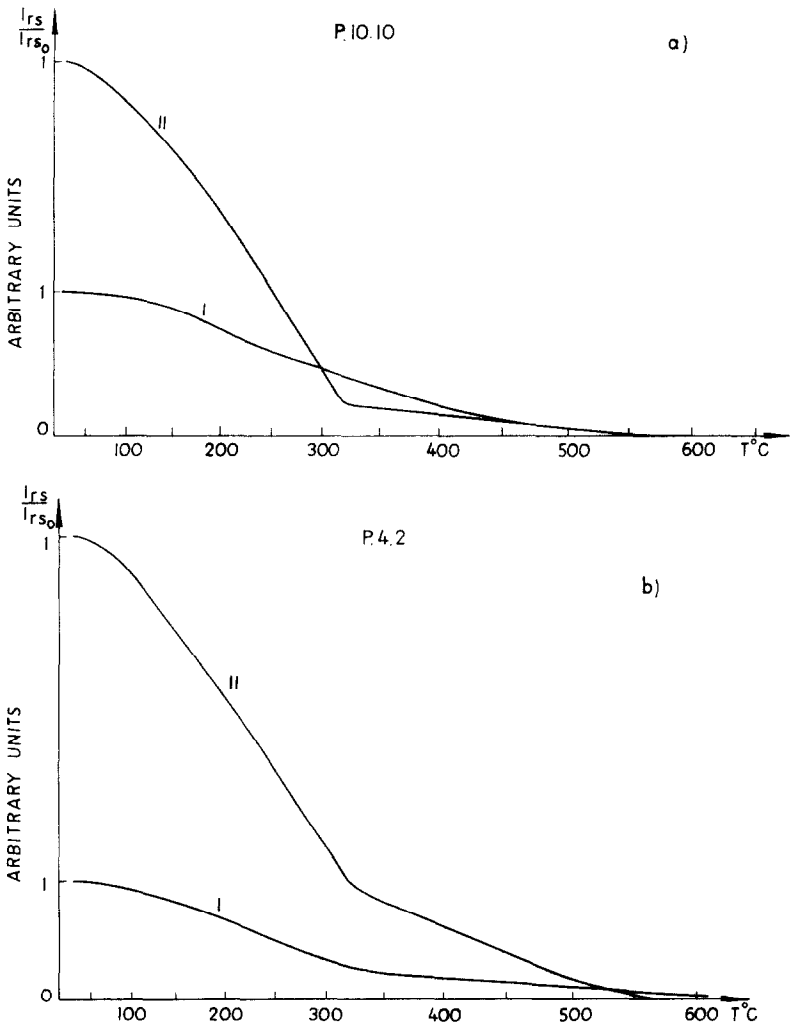


Fig. 4. Temperature dependence of isothermal saturation remanence (I_{rs}) imposed in 1.6 T for Paleozoic sediments. (a) from Bellsund; (b) from Festningen. *1* —first heating; *2* —second heating.

magnetic phase of pyrrhotite. The pyrrhotite is supposed to be generated from pyrite. Turner (1975) showed that the pyrite converts itself partly into pyrrhotite and partly into magnetite during heating. Pyrrhotite may be responsible for unstable magnetization in the Swiss Helvetic limestones (Kligfield and Channell, 1981), whereas Van der Voo and Lowrie (1979) found that the limestones of the Swiss Jura mountains, containing primary magnetite and pyrite, were suitable for paleomagnetic study. As the blocking temperatures of NRM fall into two ranges ($T_b > 400°C$ and T_b of 100°–250°C), one may assume that not the pyrrhotite, but the magnetite is the main carrier of NRM. This magnetite may be of primary origin or may be derived

TABLE 3

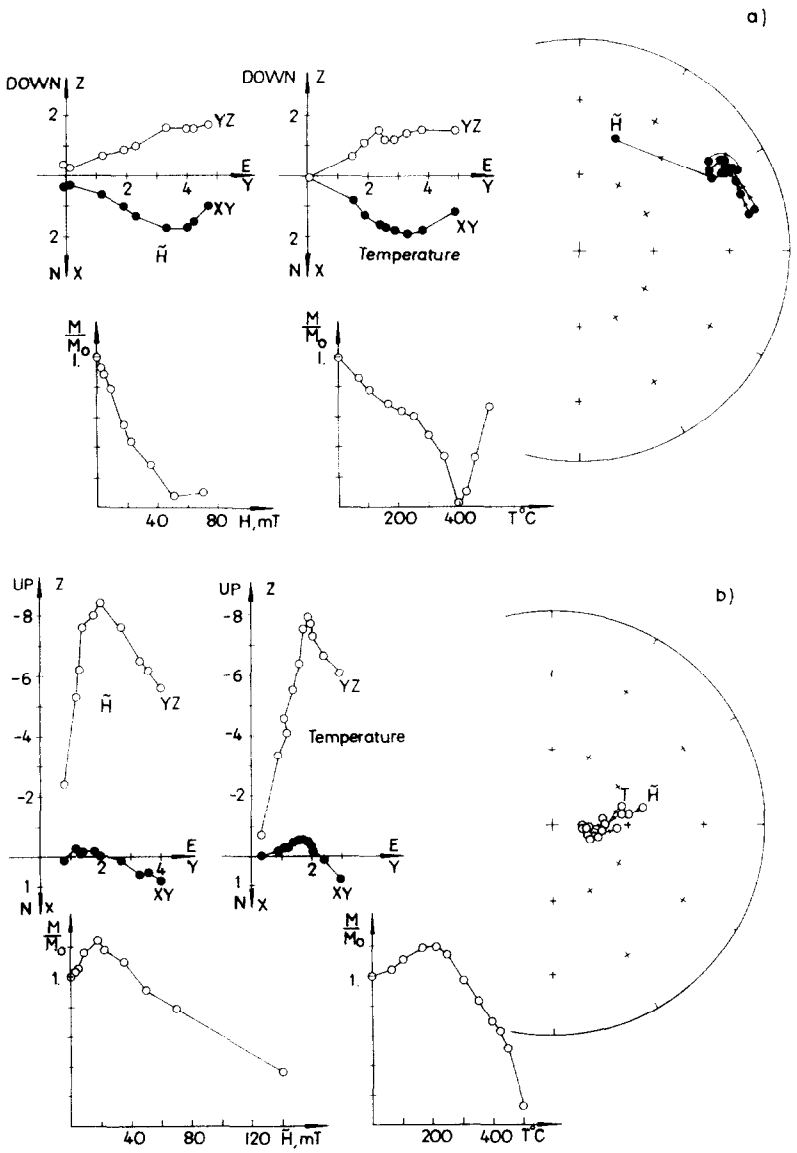
The mean directions before and after tilt corrections *

Group of directions		N	Dec. (°)	Inc. (°)	k	α_{95}	λ' (°)	ϕ' (°)	δ_p	δ_m
Hyrnefjellet formation	a	9	38.7	21.3	1.8	56.9				
shallow direction	b	9	49.5	45.5	16.4	13.1	34.4	138.9	10.6	16.6
Steep direction 1	a	7	359.4	50.8	3.9	34.8				
	b	7	353.8	55.5	24.6	12.4	48.0	201.6	12.6	17.7
Steep direction 2	a	9	12.3	87.7	4.4	27.6				
	b	9	32.0	82.6	90.4	5.4	82.3	106.9	10.4	10.6
Steep direction 3	a	7	79.4	−68.6	19.9	13.8				
	b	7	52.0	−84.7	67.6	7.4	69.7	349.7	14.4	14.6

* a—directions before tilt correction; b—directions after tilt correction; N is the number of sites.

from pyrite. Rock magnetic studies of Spitsbergen limestones are described in detail by Jeleńska and Vincenz (1984), and paleomagnetic investigations by Vincenz and Jeleńska (1984).

As in the case of the dolerites, stepwise demagnetization procedures were employed to remove secondary components of remanence. Figure 5 shows examples of typical AF demagnetization curves, Zijderveld diagrams and directional changes. It can be seen from the Zijderveld diagrams, that at least two or three components of NRM are present, and the final component seldom decreases linearly to the origin.



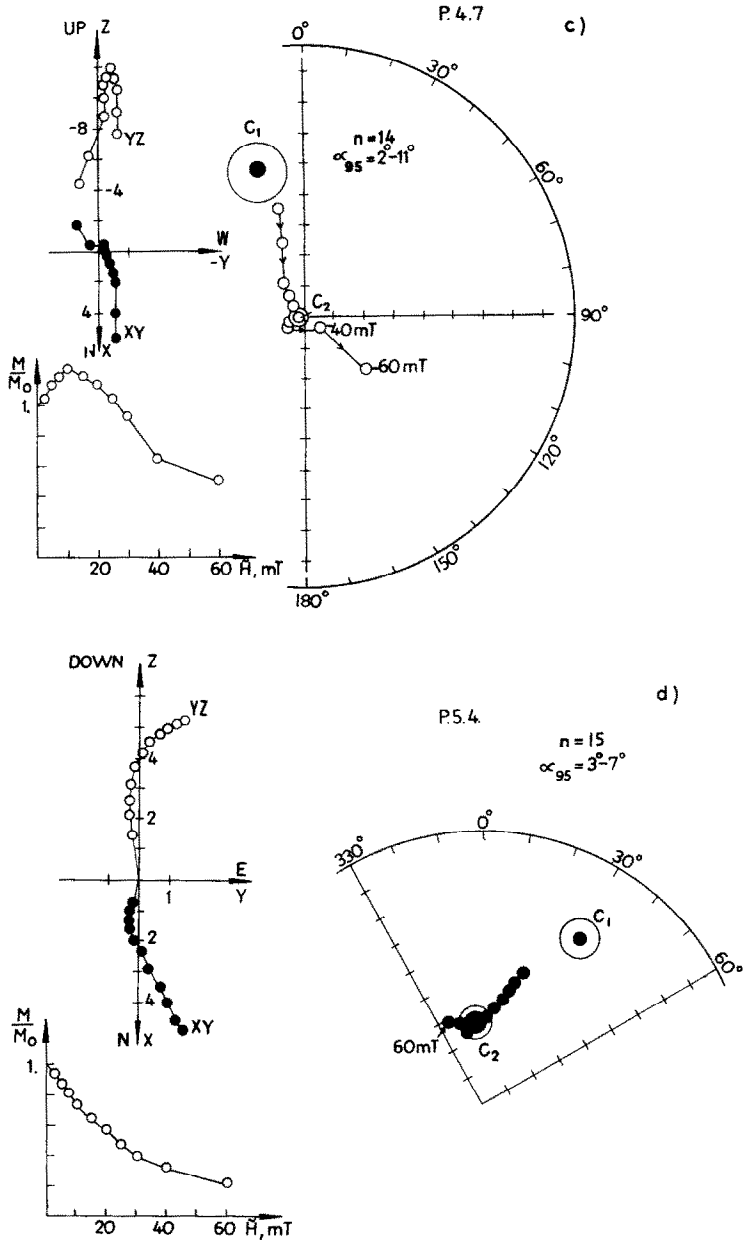


Fig. 5. Examples of demagnetization curves, Zijderveld diagrams and directional changes on stereonet for Paleozoic sediments. (a) and (b) are the curves for individual specimens, showing thermal and AF demagnetization. (c) and (d) are mean data averaged for site, demonstrating the case, when the removed component is of opposite polarity (c), and the case of components of overlapping coercivity spectra (d). C_1 and C_2 are the mean site components of NRM obtained by multicomponent analysis shown with circles of 95% confidence; n—number of specimens used to calculate the mean data; α_{95}—the range of 95% confidence circle along the demagnetization path. Units: 8×10^{-5} Am^{-1}. Legend as in Fig. 2.

Hence in order to evaluate all directions of remanence an analysis of the step demagnetization data has been applied.

For a given specimen, vector substraction and the method of intersection of demagnetization planes developed by Hofman and Day (1978) have been applied. The analysis was made by application of a computer program written by Kirschvink (1980). The analysis usually revealed two directions, one close to the beginning of the demagnetization path, and the other one close to the final section. The subtracted component of NRM is sometimes of opposite polarity. On the stereonet in Fig. 5c, d these components are marked by letters C_1 and C_2. The removed component, usually, is of a shallow down-dipping direction (C_1). The other, more stable component is of a steep inclination (C_2).

The fold test might be applied to the directions found in the limestone from Hyrnefjellet formation, which were sampled from normal and overturned layers (Table 3). The positive fold test indicates that all directions were acquired before the Tertiary, while the folding took place.

Figure 6 shows the pole positions derived from these directions together with the available paleomagnetic poles for Spitsbergen. The pole positions derived from steep directions compare favorably with different pole positions found in Mesozoic

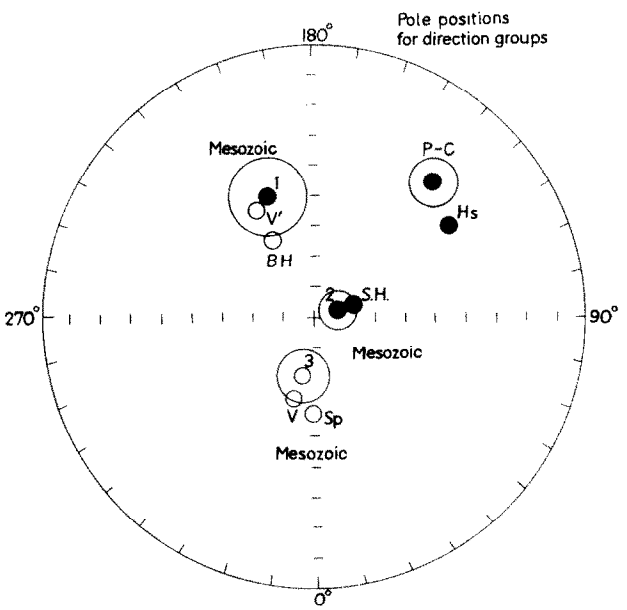


Fig. 6. Pole positions for Paleozoic sediments compared with other Permian and Late Mesozoic poles of Spitsbergen. *P–C*—1, 2 and 3 pole positions for groups of directions in Paleozoic sediments under study shown with circles of 95% confidence; *V*—Mesozoic pole (Vincenz et al., 1981); *V*—Mesozoic pole (Vincenz et al., 1981); *BH*—Mesozoic pole (Briseid and Halvorsen, 1974); *SH*—Mesozoic pole (Sandal and Halvorsen, 1973); *Sp*—Mesozoic pole (Spall, 1968; *Hs*—Permian pole (Jeleńska et al., 1978). Legend as in Fig. 3.

TABLE 4

The mean directions for formations *

Formation	n	N	Dec. (°)	Inc. (°)	k	α_{95} (°)	λ' (°)	ϕ' (°)	δ_p	δ_m
Brachiopod Cherty Limestone	117	18	46.8	35.5	11.8	10.5	27.6	143.3	7.0	12.1
Spirifer Limestone	19	3	40.7	56.8	233.4	8.1	46.1	145.8	8.5	11.7
Treskelodden Beds	40	5	56.2	43.1	12.9	22.3	31.3	132.3	17.1	27.7
Hyrnefjellet and Uppermost Treskelodden Beds	52	9	49.5	45.5	16.4	13.1	34.4	138.9	10.6	16.6
Mean pole for Brachiopod Cherty Lm., Spirifer Lm. and Treskelodden Beds		3	47.0	46.8	57.8	16.4	35.8	141.3	13.6	21.1
Mean pole for Brachiopod Cherty Lm., Spirifer Lm., Treskelodden Beds and Hyrnefjellet Beds		4	47.6	46.5	85.3	10.0	35.5	140.7	8.2	12.8

* N is number of sites; n is number of specimens in site, λ' and ϕ' are the latitude and longitude of the paleomagnetic pole in the northern hemisphere; δp and δm are the semi-axes of the oval of confidence of the pole.

dolerites. This might be explained by the fact that the sediments have been intruded at some horizons by Mesozoic dolerites and might have been remagnetized at these points. The pole position derived from shallow direction marked by letters $P-C$ was compared with the Permian pole from south Spitsbergen (Jeleńska et al., 1978).

In Table 4 mean poles for particular geological formations of Paleozoic limestones calculated from shallow directions, are listed. The positions of paleomagnetic poles are shown in Fig. 7. The shallow direction ($D = 47.6°$; $I = 46.5°$; $k = 85.3$; $\alpha_{95} = 10.0°$) might be of Permo-Carboniferous age. The positive fold test indicated that the direction was acquired before Tertiary folding. There have been no serious tectonic events in Spitsbergen since the Devonian (Harland, 1969; Birkenmajer, 1981) which might be responsible for widespread remagnetization of the Paleozoic limestones. The overprint of Mesozoic igneous events has been recognized and isolated.

The geomagnetic field obtained for the Paleozoic sediments from Spitsbergen has normal polarity. In the period between the Late Carboniferous and Late Permian the geomagnetic field had predominantly reverse polarity. The period termed the Kiaman Magnetic Interval (McElhinny, 1973) started within the Westphalian or Stephanian and lasted to the Tartasian. According to a tentative examination of polarity based on Russian data Khramov (1982) found that the transition from reversed to normal polarity occurred about 240 Ma B.P. A narrow zone of normal

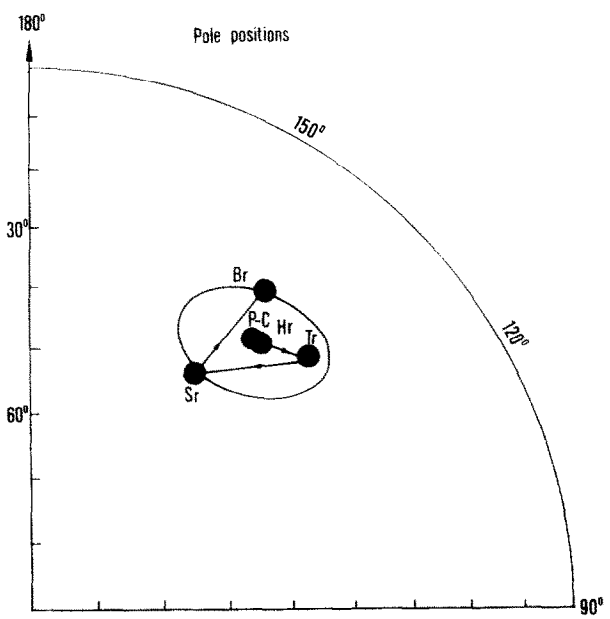


Fig. 7. Pole positions for geological formations of Paleozoic sediments. *Br*—Brachiopod Cherty Limestones fm.; *Sr*—Spirifer Limestones fm.; *Tr*—Treskelodden Beds fm.; *Hr*—Hyrnefjellet and bottom of Treskelodden fm.; *P–C*—mean Paleozoic pole with oval of 95% confidence. Legend as in Fig. 3.

polarity was also found at the Permo-Carboniferous boundary. Because the Middle Permian sediments are missing in Spitsbergen it might be that the sediments under study fall within these zones of normal polarity.

The alternative explanation which can not be excluded is the configuration of the geomagnetic field itself. Briden et al. (1970) and Westphal (1977) suggest that a large non-dipole component of the field was present during the Permian.

DISCUSSION

The Mesozoic ($\lambda' = 65.2°N$; $\phi' = 221.7°E$; $\alpha_{95} = 12.3°$) and Paleozoic ($\lambda' = 35.5°N$; $\phi' = 140.7°E$, $\alpha_{95} = 10.0°$) paleomagnetic poles for Spitsbergen are distinct from the corresponding North American and European ones (Fig. 8). It is difficult to trace the movement of Svalbard from the Permian time to present. There have probably been several displacements which have reflected global geotectonic events. These events started with the activity along the Alpha Ridge in the Arctic Basin during the Cretaceous period (Vogt et al., 1979).

The separation of Greenland and Europe proceeded through a succession of stages gradually developing the present form of the Norwegian Sea (Johnson and Heezen, 1967; Harland, 1969; Birkenmajer, 1972; Talwani and Eldholm, 1974; Birkenmajer, 1981). During the first stage of opening of the Norwegian Sea,

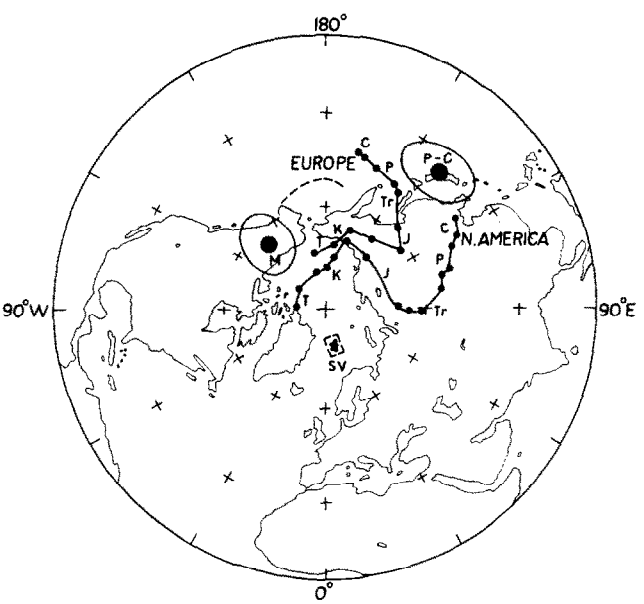


Fig. 8. Late Mesozoic paleomagnetic pole (M) and Paleozoic pole ($P-C$) for Spitsbergen with ovals of 95% confidence. Polar wander path for Europe and for North America for the time interval 20–280 Ma (Briden et al., 1981; Morel and Irving, 1981).

Greenland slid against Svalbard along a dextral fault located between northeast Greenland and Spitsbergen known as the De Geer Line (shear zone) and its present expression the Spitsbergen Fracture Zone. During the second stage the extensional processes along the mid-oceanic ridge began.

Placing Svalbard in the pre-drift position near the northeast corner of Greenland shifts the Mesozoic pole of Spitsbergen along the paleo-meridian farther from the polar wander path for Europe and North America. Both Mesozoic and Paleozoic poles would approach the North American AWP, if the simultaneous clockwise rotation by about 30° had been postulated. Even then, both poles would still lie to the south of the corresponding ones for North America and Europe.

The computed, Permo-Carboniferous paleolatitude for Svalbard of 27°N compares favorably with the paleolatitude predicted by Birkenmajer (1964), who also suggests a low latitude, between 30° and 35°N, based upon the rich, warm, coral-brachiopod fauna of the Lower Permian in Spitsbergen. The paleomagnetic data obtained by Athavale and Sharma (1974) gives about 18°N for Lower Triassic position of Central Greenland, which agrees well with the pre-drift position of Svalbard, which was about 10°–12° to the north.

According to Harland (1969) Greenland moved northword with respect to North America and Europe during the Arctic–Atlantic opening. It is possible however, that at the beginning of the process Svalbard had been pushed with Greenland to the north, and that next the dextral translation moved it again to the south.

We have to stress that the above speculations need confirmation by additional paleomagnetic data from Spitsbergen and other polar regions. Although the Mesozoic pole seems to be well documented, the Permo-Carboniferous data are controversial. However, the discussion based mainly on the Mesozoic data leads to the conclusion that Svalbard has moved independently from Europe in the Late Mesozoic and behaved as a mobile block or even microplate. The same conclusion was drawn recently by Savostin and Karasik (1981). They examined a kinematic model for lithospheric plate motion in the Arctic region, and considered the recent tectonics of the Arctic Basin and northeastern Asia as a result of interaction between three plates: North America, Eurasia and Svalbard. They concluded that the Svalbard microplate has existed at least for the last 60 Ma.

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